Bank Customer Churn Prediction

Data Analysis & Machine Learning Project

Abdallah Mohamed Draz

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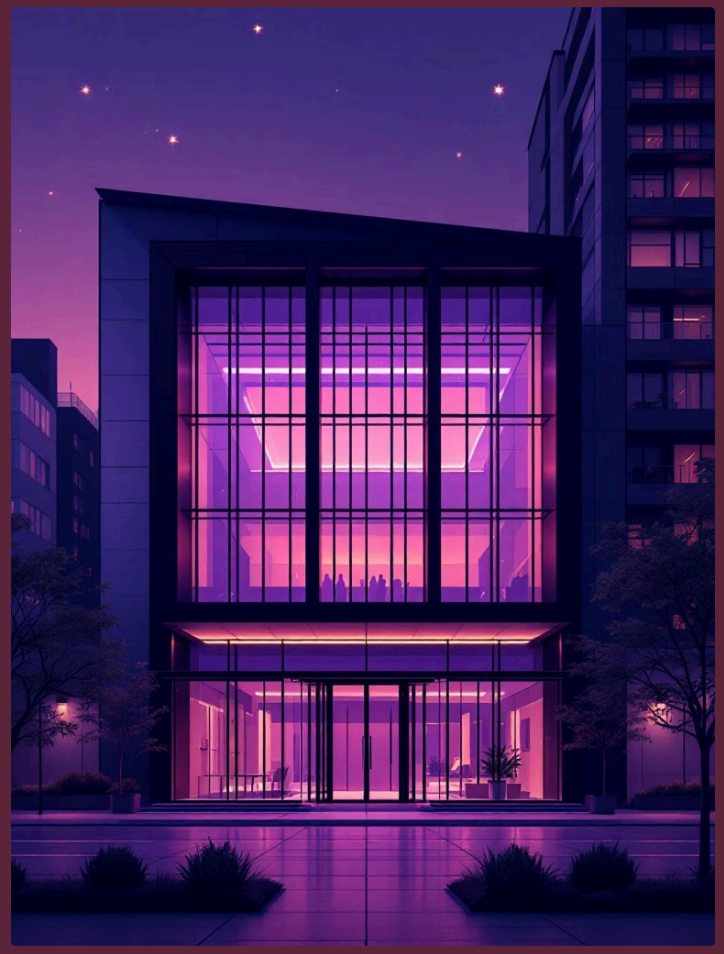
Project Overview



Problem Statement

Banks lose valuable customers to competitors every year. Identifying customers likely to leave allows banks to proactively reduce churn through targeted retention strategies.

Project Goal

Build a comprehensive analysis framework to predict customer churn, understand behavioral patterns, and provide actionable business insights to improve customer retention rates.

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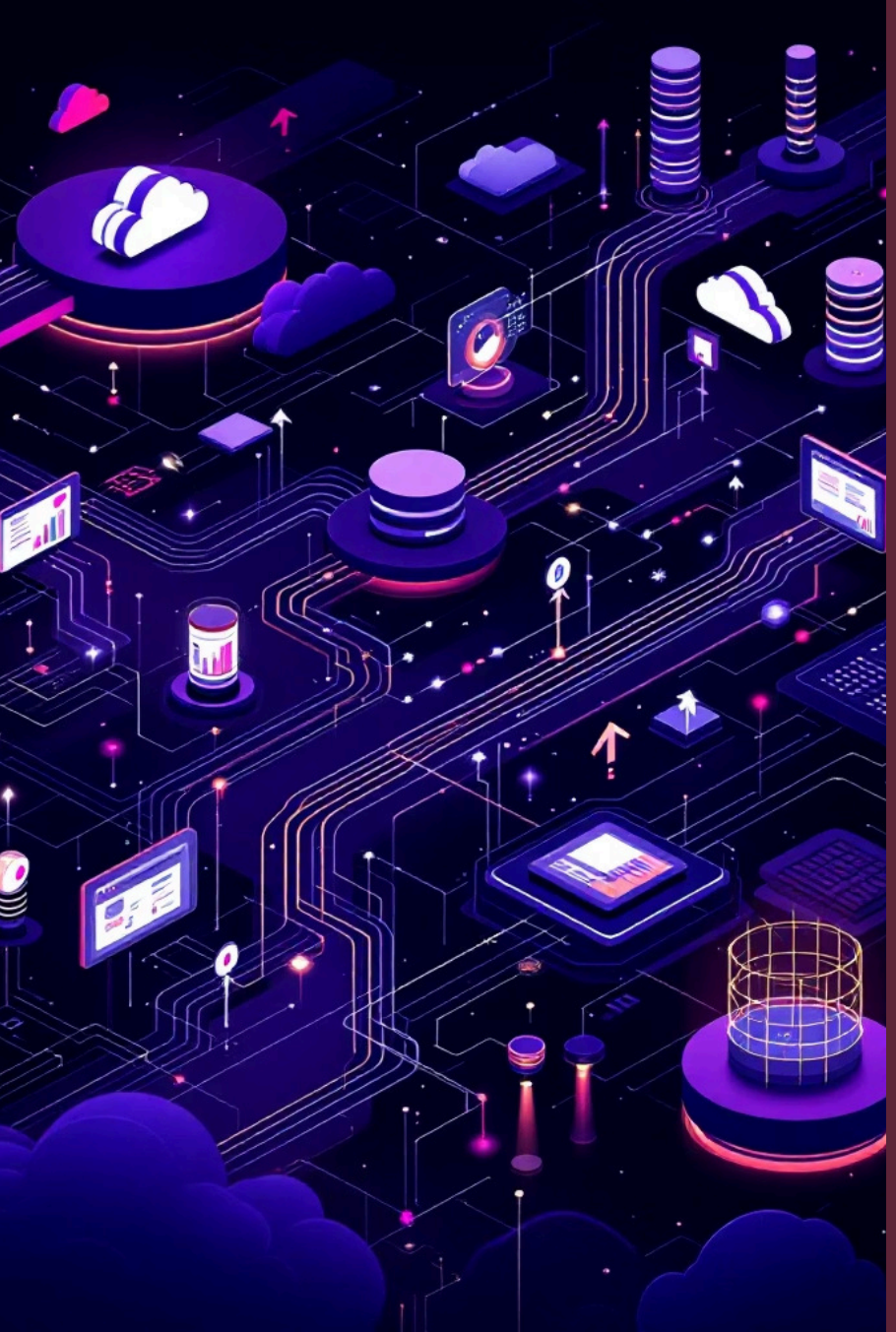

Dataset & Methodology

Dataset Overview

10,000+ customer records with 10+ features including demographics, account details, and engagement metrics

Tools Used

Python (Pandas, NumPy), Power BI for visualization, statistical analysis techniques

Data Preparation

Missing value imputation, categorical encoding, feature engineering, normalization applied where necessary

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Key Performance Indicators

10K+

Total Customers

Complete customer base analyzed

68%

Active Customers

Currently engaged with bank services

20.4%

Churn Rate

Customers who left in past year

$76,500

Avg. Balance

Mean account balance across customers

1.5

Products per Customer

Average number of bank products used

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Customer Demographics

Geographic Distribution

France: 5,014 customers (50%)
Germany: 2,509 customers (25%)
Spain: 2,509 customers (25%)

Gender Split

Male: 5,457 customers (54.6%)
Female: 4,543 customers (45.4%)

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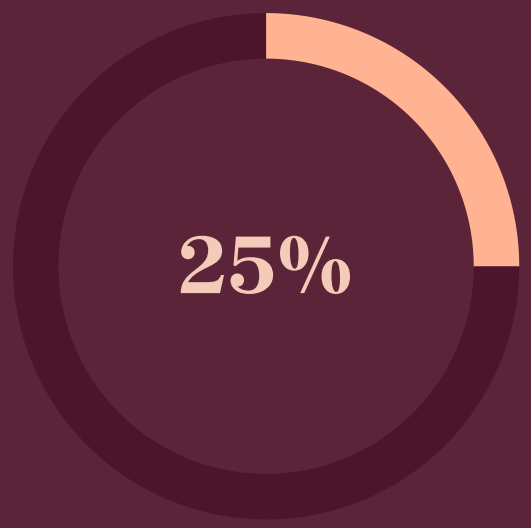
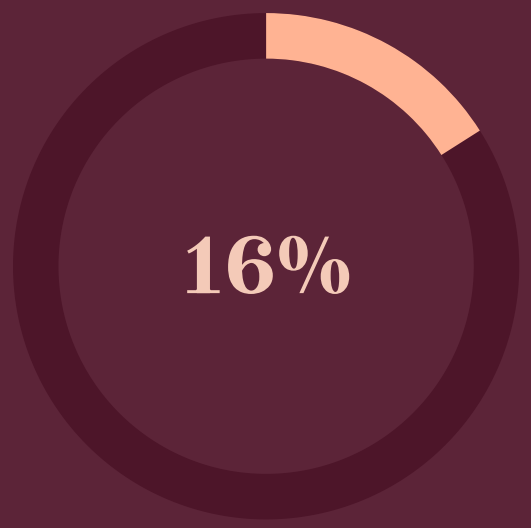
Churn Analysis by Segment



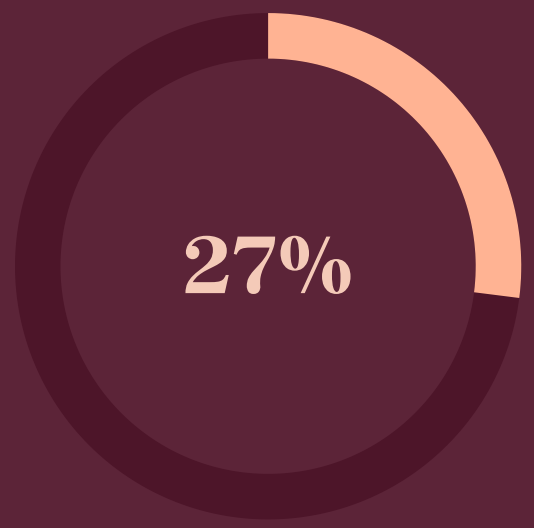
Female Churn Rate

Higher than male customers



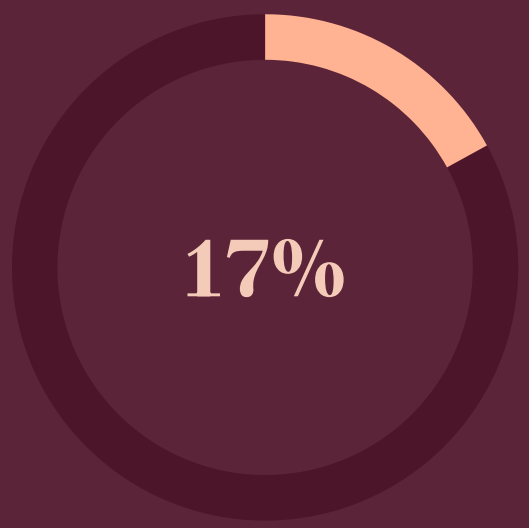
Male Churn Rate

Lower attrition among males



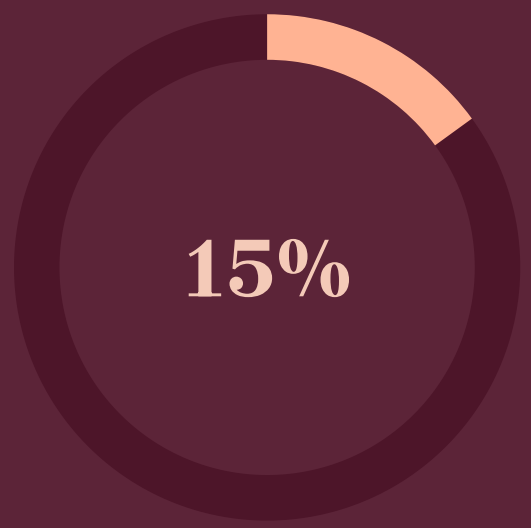
Germany Churn

Highest regional attrition



France Churn

Mid-range churn rate



Spain Churn

Lowest regional attrition

Key Finding: Gender and geography are significant predictors of churn behavior, with females in Germany showing highest risk.

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Financial Insights

Balance & Churn Relationship

Customers with lower account balances show higher churn rates, suggesting balance is a key indicator of customer loyalty.

Credit Score Impact

Higher credit scores correlate with lower churn rates, though the relationship is not perfectly linear across all score ranges.

High-Value Customers

Top 10% by balance account for 45% of total deposits, making their retention critical to revenue stability.

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Key Insights & Business Recommendations

Target High-Risk Segments

Focus retention efforts on female customers in Germany with lower balances and fewer products

Enhance Engagement

Proactive outreach to inactive members through personalized offers and improved service experiences

Product Bundling

Encourage adoption of multiple products to increase customer stickiness and reduce attrition risk

Data-Driven Decision Making: These insights enable proactive customer retention strategies rather than reactive responses to churn events.

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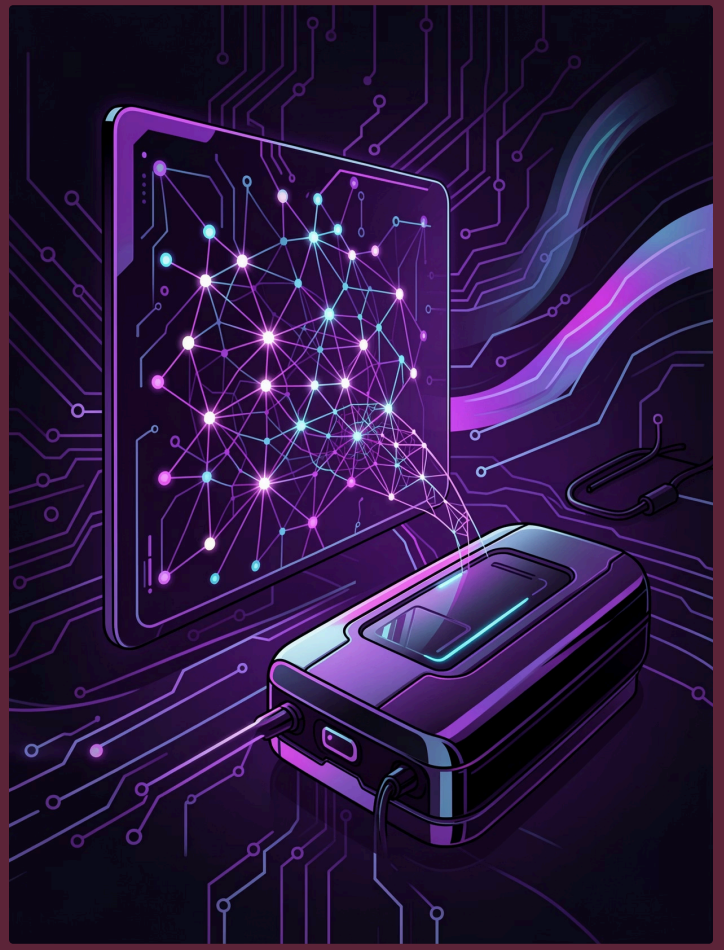
Conclusion & Future Work

Project Summary

This analysis successfully identified key churn drivers including gender, geography, account balance, and product usage. The framework provides actionable insights for targeted retention campaigns and improved customer engagement strategies.

Next Steps

- Build and train machine learning models for churn prediction
- Deploy predictive model into production environment
- Develop real-time dashboard with automated alerts
- Implement A/B testing for retention strategies
- Monitor and measure impact of interventions



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Thank You

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